

**MIDTERM EXAMINATION**  
**SEMESTER FALL 2004**  
**MTH401- Differential Equations**

**Total Marks: 50**  
**Duration: 60min**

**Instructions**

1. Attempt all questions.
2. The Time allowed for this paper is 60 minutes.
3. This examination is closed book, closed notes, closed neighbors; any one found cheating will get zero grades in the course MTH401 Differential Equations.
4. You are not allowed to use any type of Table for Formulae of Differentiation and integration during your exam.
5. Each objective type question carries 2 marks and each Descriptive question carries 10 marks. So write every step of the solution of descriptive question to get maximum marks.
6. Do not ask any questions about the contents of this examination from anyone. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.

A differential equation is said to be ordinary differential equation if it contains only ordinary derivatives with respect to single variable.

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A solution of the differential equation of the form  $y = f(x)$  is called the implicit solution.

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ogistic equations are applications of non-linear equations.

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**Note: Solve these papers by yourself**  
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The given functions  $f_1(x) = 5$ ,  $f_2(x) = \cos^2 x$ ,  $f_3(x) = \sin^2 x$  are linearly independent.

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set of functions whose wronskian is zero then set of functions may or may not be dependent.

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(a)

Define separable form. Just separate the variables of the given differential equation.

The differential equation of the form  $dy/dx = f(x, y)$  is called **separable** if it can be written in the form  $dy/dx = h(x)g(y)$

$$\sec y \frac{dy}{dx} = \sin x - y \quad \sin x \quad y$$

$$\sec y \frac{dy}{dx} = \sin x \quad y - \sin x - y$$

$$\sec y \frac{dy}{dx} = \sin x \quad y - \sin x - y$$

$$\sec y \frac{dy}{dx} = \sin x \cos y \quad \cos x \sin y - \sin x \cos y \quad \cos x \sin y$$

$$\sec y \frac{dy}{dx} = 2 \cos x \sin y$$

$$\frac{dy}{dx} = \cos x \sin y$$

$$\frac{dy}{\sin 2y} = \cos x dx$$

$$\csc 2y dy = \cos x dx$$

(b) Check whether the given differential equation is exact or not if not then make it exact also show that it is exact (

$e^x dx + \frac{1}{\cos y} - 2y \csc y dy = 0$   
Just make the equation exact do not solve it further)

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$$e^x dx - e^x \cot y - 2y \cos ecy dy = 0$$

$$M_{x,y} = e^x, N_{x,y} = e^x \cot y - 2y \cos ecy$$

$$M_y = 0, N_x = e^x \cot y$$

$$M_y \neq N_x$$

Thus it is not exact now we apply techniques to make it exact

$$e^x dx - e^x \cot y - 2y \cos ecy dy = 0$$

$$N_x - M_y = e^x \cot y - 0 = e^x \cot y$$

$$I.F. = e^{\int \cot y dy} = e^{\ln \sin y} = \sin y$$

$$e^x \sin y dx - e^x \cos y - 2y dy = 0$$

$$M_{xy} = e^x \sin y, N_{xy} = -e^x \cos y - 2$$

$$M_y = e^x \cos y, N_x = e^x \cos y$$

$$M_y = N_x$$

Which shows that equation is exact

(a) Solve Bernoulli equation  $y - 2 \frac{dy}{dx} = y^3(x-1)$  ( )

Just make the given equation linear in v, do not integrate

$$xy - \frac{dy}{dx} = y^3(x-1)$$

$$xy^{-2} - \frac{dy}{dx} = y^{-3}(x-1)$$

$$\text{put } y^{-2} = v$$

$$-2 \frac{dy}{dx} y^{-3} = \frac{dv}{dx}$$

$$-\frac{dy}{dx} y^{-3} = \frac{1}{2} \frac{dv}{dx}$$

Then

$$\frac{1}{2} \frac{dv}{dx} = vx - (x-1)$$

$$\frac{dv}{dx} = 2vx - 2(x-1)$$

(b) The half-life of a radioactive isotope of the lead, Pb-209, is 3.4 hours. If a sample of 2 grams of the lead is present initially, how long will it take for 80% of the lead to decay?

Further Just make the model of the radioactive decay as well as describe the given conditions do not solve

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Suppose that  $A_0$  is the initial amount of isotope, as given  $A_0$  is 100 and  $A(t)$  be the amount present at time  $t$  it is governed by the differential equation.

$$\frac{dA}{dt} = -kA$$

Integrate both sides

In  $Aktc$

$$Ae^{-kt} = c$$

$$A = ce^{kt}$$

$$A = P_0 e^{kt} \quad \text{say } A_0 = e^c$$

Where  $A_0$  is the initial amount of isotope

$$A_0 = 2 = A(0)$$

$$A(4) = 2/2 = 1$$

Then we have to find time when radioactive isotope will take 80% decay. So as  $A$  initially given 2 and 80% of 2 is 8/5 so decay would be  $1 - 8/5 = -3/5$

$$P(t) = -3/5, \quad t = ?$$

(a)

Find a second solution of following differential equations where the first solution is given by  $y_1 = x^2$  (also write the formulae).

$$x^2 y'' - 4xy' + 6y = 0; \quad y_1 = x^2$$

differential equation can be written as

$$y'' - \frac{4}{x} y' + \frac{6}{x^2} y = 0$$

the 2<sup>nd</sup> solution is given by

$$y_2 = y_1 \int \frac{e^{-\int p dx}}{y_1^2} dx$$

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$$y_2 = x^2 \int e^{-\int \frac{4}{x^2} dx} \frac{1}{x^2} dx$$

$$y_2 = x^3 \ln x \int \frac{e^{\int \frac{4}{x} dx}}{x^3 \ln x} dx$$

$$y_2 = x^2 \int \frac{e^{4 \ln x}}{x^4} dx$$

$$y_2 = x^3 \ln x \int \frac{e^{\ln x^4}}{x^4} dx$$

$$y_2 = x^2 \int \frac{x^4}{x^4} dx$$

$$y_2 = x^2 \int dx$$

$$y_2 = x^3$$

(b)

Solve the differential equation by the undetermined coefficient

(superposition approach)

If complimentary solution is given below

$$y_c = c_1 e^{3x} + c_2 e^{-x}$$

Then just find particular solution.

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e find a particular solution of non-homogeneous differential equation.

$$y_p = A \cos \theta + B \sin \theta$$

Then

$$y'_p = -A \sin \theta + B \cos \theta$$

$$y''_p = -A \cos \theta - B \sin \theta$$

Substituting in the given differential equation

$$-A \cos \theta - B \sin \theta - 2(-A \sin \theta + B \cos \theta) = 3A \cos \theta + B \sin \theta + 4 \sin \theta$$

$$-A - 2B - 3A \cos \theta - B = 2A - 3B \sin \theta + 4 \sin \theta$$

From the resulting equations

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$$\begin{aligned}
 -A - 2B - 3A &= 0; & -B + 2A - 3B &= 4 \\
 -4A - 2B &= 0; & 2A - 4B &= 4 \\
 2AB &= 0; & AB + 2 &\rightarrow A + 2 + 2B \\
 \rightarrow 2(2 + 2B) + B &= 0 \\
 \rightarrow 4 + 5B &= 0 \\
 \rightarrow B &= -\frac{4}{5} \\
 \rightarrow A &= -\frac{2}{5} \\
 y_p &= \frac{2}{5} \cos \theta - \frac{4}{5} \sin \theta
 \end{aligned}$$

(a)

(annihilator operator).

Solve differential equation by the undetermined coefficient  
 $\frac{d^2y}{dx^2} - 4 \frac{dy}{dx} = e^{2x}$

If complimentary solution is given below

$$y_c = c_1 + c_2 e^{4x}$$

Then just find general solution.

In this case of input function is

$$g(x) = e^{2x}$$

further

$$(D - 2)(g(x)) = (D - 2)(e^{2x}) = 0$$

Therefore the differential operator  $D^3$

$$(D - 2)(D^2 - 4D)y = (D - 2)(e^{2x})$$

annihilates the function g. operating on both sides

$$(D - 2)(D^2 - 4D)y = 0$$

This is the homogeneous equation of order 3. Next we solve this higher order equation.  
 Thus auxiliary equation is

$$(m - 2)(m^2 - 4m) = 0$$

$$m(m - 2)(m - 4) = 0$$

$$m = 0, 2, 4$$

Thus its general solution of the differential equation must be

$$y = c_1 + c_2 e^{2x} + c_3 e^{4x}$$

(b)

Solve the differential equation by the variation of parameters

Note: Solve these papers by yourself

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$$y'' - 4y = 3 \cos x$$

If complimentary solution is given below

$$y_c = c_1 e^{3x} + c_2 e^x$$

Then just find particular solution **(do not integrate)**.

$$y'' - 4y = 3 \cos x$$

This equation is already in standard form

$$y'' + P(x)y' + Q(x)y = f(x)$$

Therefore, we identify the function  $f(x)$  as

$$f(x) = 3 \cos x$$

We construct the  $y_1, y_2 = e^x$  so

determinants

$$W(y_1, y_2) = \begin{vmatrix} e^{3x} & e^x \\ 3e^{3x} & e^x \end{vmatrix} = e^{3x}e^x - 3e^{3x}e^x = -2e^{4x}$$

$$W_1 = \begin{vmatrix} 0 & e^x \\ \cos x e^x & e^x \end{vmatrix} = 0 \cdot e^x - \cos x e^x e^x = -\cos x e^{2x}$$

$$W_2 = \begin{vmatrix} e^{3x} & 0 \\ 3e^{3x} & \cos x \end{vmatrix} = e^{3x} \cos x - 0 = \cos x e^{3x}$$

$u_1$  and  $u_2$

We determine the derivatives of the

$$u_1 = \frac{W_1}{W} = \frac{-\cos x e^{2x}}{-2e^{4x}} \rightarrow u_1 = \frac{\cos x}{2} e^{-2x}$$

$$u_2 = \frac{W_2}{W} = \frac{\cos x e^{3x}}{-2e^{4x}} \rightarrow u_2 = -\frac{\cos x}{2} e^{-x}$$

$$y_p = u_1 e^{3x} + u_2 e^x$$

is required particular solution

**Note:** Solve these papers by yourself

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